



New & Delete ÷

new is a **memory allocation operator** used to dynamically allocate memory on the **heap**.

syntax : **new** data-type

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Program: Allocating a Single Variable ÷

```
#include <iostream>
```

```
using namespace std;
```

```
int main()
```

```
{
    cout << "robustab A 220/2 most allot n/">>>trn
```

```
    int *ptr = new int // Allocate memory for an integer
                        // and return the address of that
                        // memory which is stored in ptr
```

```
    *ptr = 42; // Assigning value at that address
```

```
    cout << "Value : " << *ptr << endl;
```

```
    return 0;
```

```
}
```

Program: Allocating a Single Object Using **new**

```
#include <iostream>
using namespace std;
```

```
class student
{
```

```
public:
```

```
    string name;
```

```
    int age;
```

```
    Student(string n, int a)
{
```

```
    name = n;
```

```
    age = a;
```

```
}
```

```
void display()
```

```
{
```

```
    cout << "Name : " << name << " Age : " << age;
```

```
}
```

```
};
```

```
int main()
```

```
{
```

```
    Student *s1 = new Student("Alice", 20);
```

↳ **new** allocates **memory block** in **heap** for all attributes name and age.

Then the **constructor** Student("Alice", 20) is called.